DATA ANALYSIS PROJECT REPORT

Customer Purchase Behavior and Revenue Trends in Online Retail Transactions

Item	Details
Project Title	Customer Purchase Behavior and Revenue Trends in Online Retail Transactions
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Course / Section	PROGRAMMING IN PYTHON [B]
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Date of Submission	**02/05/26**

1. Executive Summary

This project analyzes customer purchase behavior and revenue trends using the UCI Online Retail dataset. The goal is to understand revenue changes over time, identify the strongest countries and products, and examine unusual high-value invoices. Pandas was used for loading, cleaning, grouping, and aggregation; NumPy was used for percentile-based classification and z-score style numerical computation; and Matplotlib was used for visual evidence. After cleaning, the analysis-ready dataset contained 524,878 positive-sales rows. Key findings show that the United Kingdom contributed 84.59% of cleaned revenue, 2011-11 was the highest revenue month with £1,503,866.78, and the IQR method identified 1,811 high-value invoice outliers.

2. Introduction and Project Objectives

Item	Content
Dataset name	Online Retail
Dataset source	UCI Machine Learning Repository: Online Retail dataset
Main project goal	Analyze online retail transactions to understand revenue trends, country/product contribution, customer purchase behavior, and high-value outliers.
Research Question 1	How does online retail revenue change over time?
Research Question 2	Which countries and products contribute most to total revenue?
Research Question 3	What purchase patterns exist, and which transactions look unusually large?

3. Dataset Description

Each row represents a transaction line item. The original columns are InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The data includes numerical, categorical, text, and date-time variables, so it is suitable for trend analysis, subgroup comparison, feature engineering, and anomaly detection.

Property	Value
Rows	541,909 raw rows; 524,878 cleaned positive-sales rows
Columns	8 original columns
Data types	Numerical, categorical, text, and date-time
Key variables	quantity, unit_price, total_revenue, invoice_month, country, product description, customer_id, invoice_no
Initial issues	Duplicates, cancelled/returned transactions, invalid quantity or price values, missing customer IDs, and inconsistent descriptions

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Inconsistent column names	Renamed to lowercase snake_case.	Improves readability and avoids coding errors.	src/cleaning.py
Wrong/unsafe data types	Converted dates and numeric fields.	Required for time-series and revenue calculations.	src/cleaning.py
Duplicate records	Removed exact duplicates.	Prevents overstated sales and counts.	src/cleaning.py
Cancelled/returned/invalid sales	Excluded cancelled invoices and non-positive quantity/price rows.	Keeps the main analysis focused on completed positive sales.	src/cleaning.py
Product description issues	Filled missing descriptions and cleaned spacing.	Improves product-level grouping.	src/cleaning.py

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
total_revenue	quantity × unit_price	Measures line-level revenue for

		totals, rankings, and trends.
invoice_month/year/day/hour	Extracted from invoice_date	Supports monthly trends and time-based behavior analysis.
market_group	UK vs International using country	Supports domestic vs international subgroup comparison.
day_type	Weekday or Weekend from invoice_date	Supports timing-based customer behavior comparison.
basket_value / basket_distinct_items	Grouped by invoice_no	Supports invoice-level purchase behavior and relationship analysis.
basket_value_category / line_revenue_zscore	Created using NumPy percentiles and z-score logic	Supports segmentation and outlier analysis.

6. Analysis Methodology

The analysis used Pandas for filtering, grouping, ranking, aggregation, and summary tables; NumPy for percentiles, condition-based features, and z-score style anomaly support; and Matplotlib for visualizations. The project included trend analysis, country/product ranking, subgroup comparison, relationship analysis, and outlier analysis. The cleaned dataset produced total positive sales revenue of £10,642,110.80, 19,960 unique invoices, and an average order value of £533.17.

7. Analysis and Visualizations

RQ1 found that 2011-11 was the strongest month with £1,503,866.78. RQ2 found that the UK dominated revenue with 84.59% and the highest revenue product was DOTCOM POSTAGE (£206,248.77). RQ3 found a positive but weak-to-moderate relationship between distinct products and invoice value (correlation 0.265); the IQR method found 1,811 high-value invoice outliers.

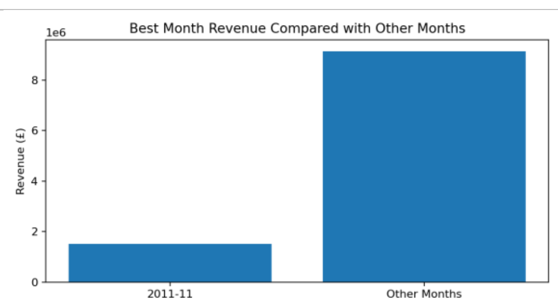

Figure 1. Best revenue month: 2011-11 vs other months.

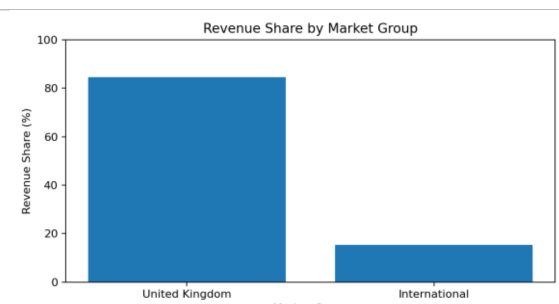

Figure 2. UK vs international revenue share.

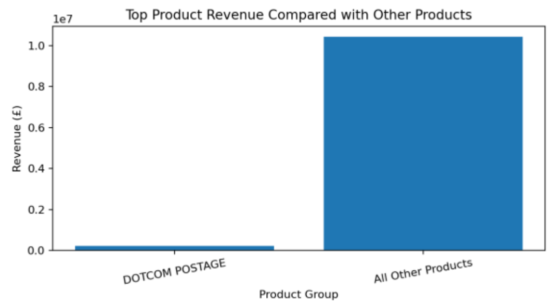

Figure 3. DOTCOM POSTAGE vs all other products.

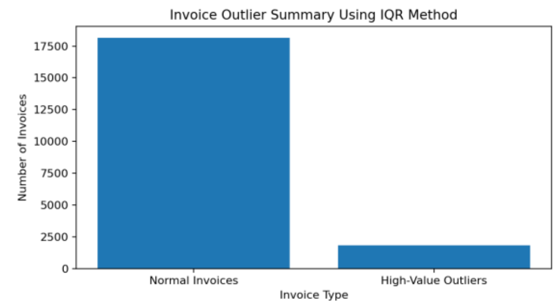

Figure 4. Normal invoices vs high-value outliers.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Percentile-based basket categories and z-score style line revenue anomaly calculation.
Purpose	Classify invoice value levels and support unusual transaction detection.
Formula / logic	z = (x − mean) / standard deviation; percentiles divide basket values into Low, Medium, and High groups.
Result / takeaway	The IQR method identified 1,811 high-value invoice outliers, and the engineered basket features supported purchase behavior analysis.

9. Key Findings

- The cleaned analysis-ready data contained 524,878 positive-sales rows, which provided enough records for meaningful business analysis.
- Total positive sales revenue was £10,642,110.80, based on 19,960 unique invoices and an average order value of £533.17.
- The United Kingdom was the strongest revenue market, contributing 84.59% of cleaned revenue.
- The highest revenue month was 2011-11, with revenue of £1,503,866.78, showing a strong late-year revenue peak.
- The highest revenue product was DOTCOM POSTAGE, with revenue of £206,248.77.
- The relationship between distinct products per invoice and invoice value was positive but not strong, with correlation 0.265.
- The IQR method identified 1,811 high-value invoice outliers, suggesting a meaningful group of unusually large purchases.

10. Limitations

- The dataset covers a limited transaction period from December 2010 to December 2011, so the findings may not represent long-term business behavior.
- Some customer IDs are missing, which limits customer-level analysis and prevents a complete customer segmentation study.
- The business is UK-based, so the country-level results are naturally biased toward the United Kingdom.
- Cancelled and returned transactions were excluded from the main positive-sales analysis; this makes revenue trends clearer but does not fully analyze refund behavior.
- The correlation between basket size and invoice value does not prove causation. Other factors, such as product price and customer type, may also influence invoice value.

11. Conclusion

This project used Pandas, NumPy, and Matplotlib to analyze online retail transactions. The results show that revenue is strongly concentrated in the UK market, that November 2011 is the strongest revenue month, and that some invoices are unusually high in value. The engineered revenue, time, market, basket, and anomaly features helped answer the research questions and support evidence-based findings.

12. References / Dataset Source

Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5BW33

Dataset page: https://archive.ics.uci.edu/dataset/352/online+retail

13. Final Submission Checklist

Checklist Item	Status
Dataset has at least 1000 rows and 6 columns	Completed
Pandas, NumPy, and Matplotlib used meaningfully	Completed
At least 3 cleaning/preprocessing steps included	Completed
At least 2 engineered features created	Completed
At least 6 analysis operations, subgroup comparisons, relationship analysis, and outlier analysis included	Completed
At least 4 visualizations included and interpreted	Completed
Findings, limitations, and conclusion included	Completed